Software Engineering & Astrophysics

Experiences in Academia & Industry



Nayera Abdessalam

TIMESTEP Internship Symposium
Student Keynote
September 2, 2026



My Background



University of Arizona

Majors: Astronomy & Physics
Minors: Computer Science & Mathematics

Data Center Internship

Academic Research

TIMESTEP Astronomical Software Engineering

IBM

Software Developer Internship

TIMESTEP's Astronomical Software Engineering Internship

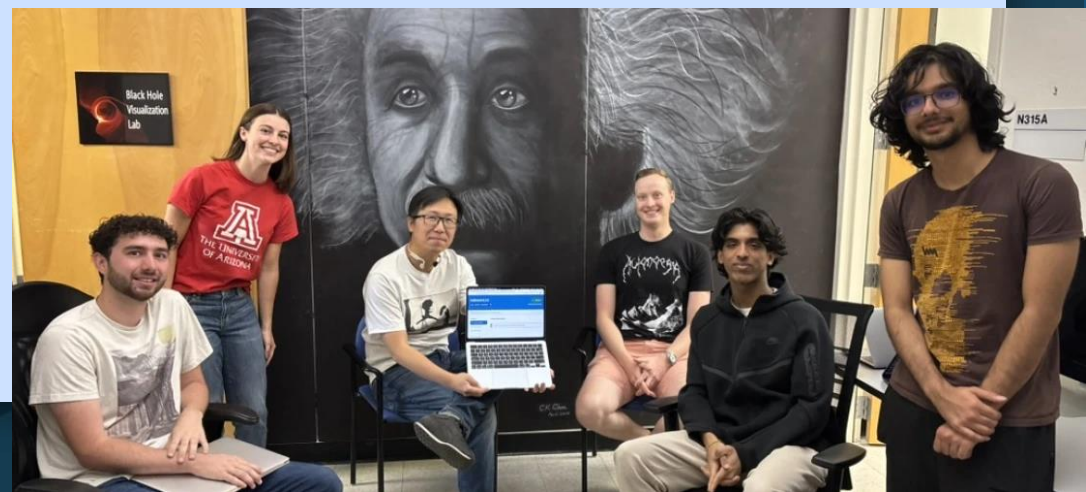
TIMESTEP

Goals:

The TIMESTEP Astrophysics Software Engineering Internship places juniors and seniors with backgrounds in astronomy and physics into software engineering teams that are working on some of the most cutting edge challenges in the field.

STEM students work together in teams of 3-4 interns to develop software to solve space sciences problems (e.g., large data sets, slow pipelines).

Student outcome: Provide a career pathway for STEM students into industry as a Software Engineer and/or graduate school for research



2025-2026 Pilot Astro SE Internship: EHT Pipeline working with Dr. CK Chan

We helped develop a modern, scalable software ecosystem for managing, inspecting, and processing Very Long Baseline Interferometry (VLBI) data for the EHT. We continued to improve the code base and add features according to what would be most useful to the scientists.

pypi.org/project/hallmark/

hallmark 0.2.0

Reproducibility is the hallmark of the scientific method

```
pip install hallmark
```

Released: May 1, 2026

LATEST RELEASE



What's unique about the Astro SE Internship

- Set me up for my IBM software engineering internship by introducing these skills to me and giving me the platform to develop them throughout the academic year



GitHub project management



Coding at industry level



Agile workflow



AI in code development



Full test suites



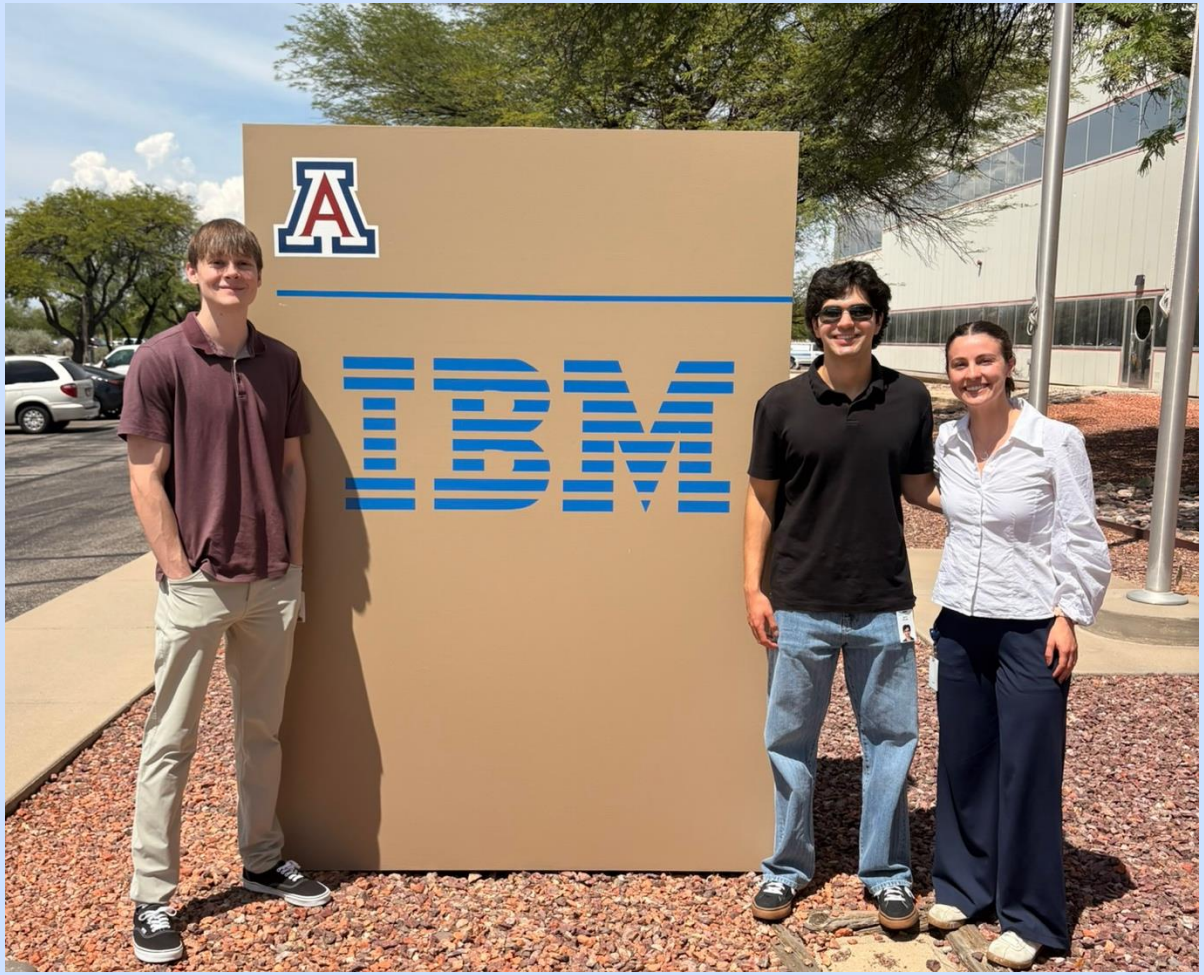
Teamwork

My summer with the IBM Storage CTO's Office



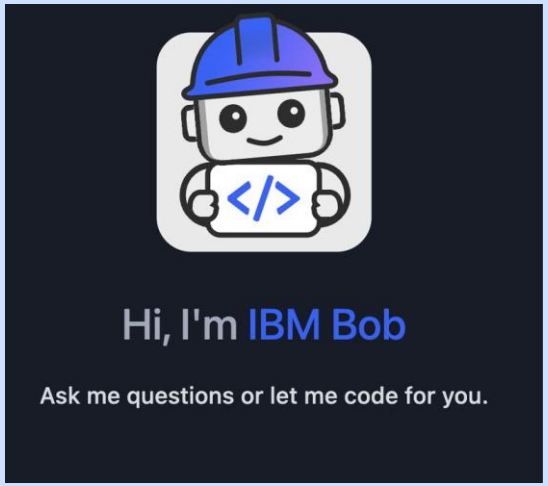
IBM®

Different backgrounds bring a greater variety of skills to a team

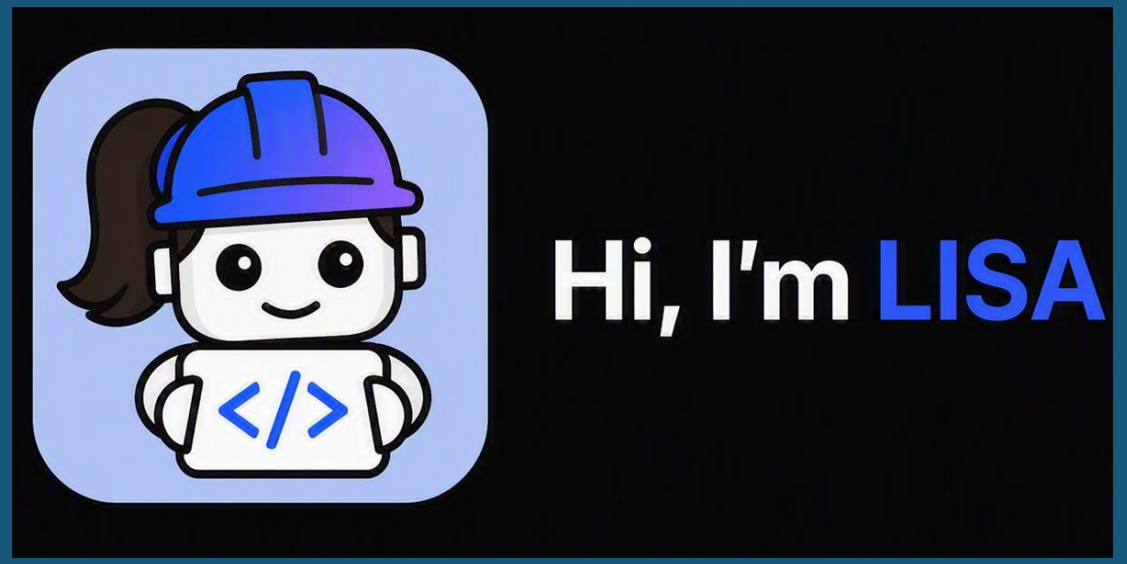


Zeb Moffat (Software Engineering/Data Science), Seth Rojas (Computer Science), and me!

Using IBM Bob* to create LISA



* Enabled by my background in Software Engineering & Python



LLM Infrastructure Sizing Agent

- Problem: Companies want to build internal AI agents/systems but don't know the infrastructure or setup it requires
- Solution: A natural language assistant that understands a range of user inputs and produces output metrics and recommendations based on complex simulation data

Multi turn conversation ···

Okay let's run nvidia/Llama-3.1-70B-Instruct-FP8 on 4 H100 SXM5 GPUs with 128 GB CPU memory and 50 TB of Storage Scale for a heavily used HR assistant.

LISA

Simulation Overview

SIMULATION	GPU	MEMORY	TP	WORKERS	KV CACHE TIERS	MODEL
Run a heavily used HR assistant RAG simulation of nvidia/Llama-3.1-70B-Instruct-FP8 on 4 H100_SXM5 GPUs (tensor_parallel_size=4, num_workers=1) with 128 GB CPU memory and 51200 GB DFS storage.	H100 SXM5	80 GB	4	1	CPU Memory, Distributed File System	nvidia/Llama-3.1-70B-Instruct-FP8

KV Cache Tier Specifications:

TIER	CAPACITY	BANDWIDTH	LATENCY
CPU Memory	128 GB	200 GB/s	200 ns
Distributed File System	51,200 GB	350 GB/s	100,000 ns

1. Executive Summary

Send a Message

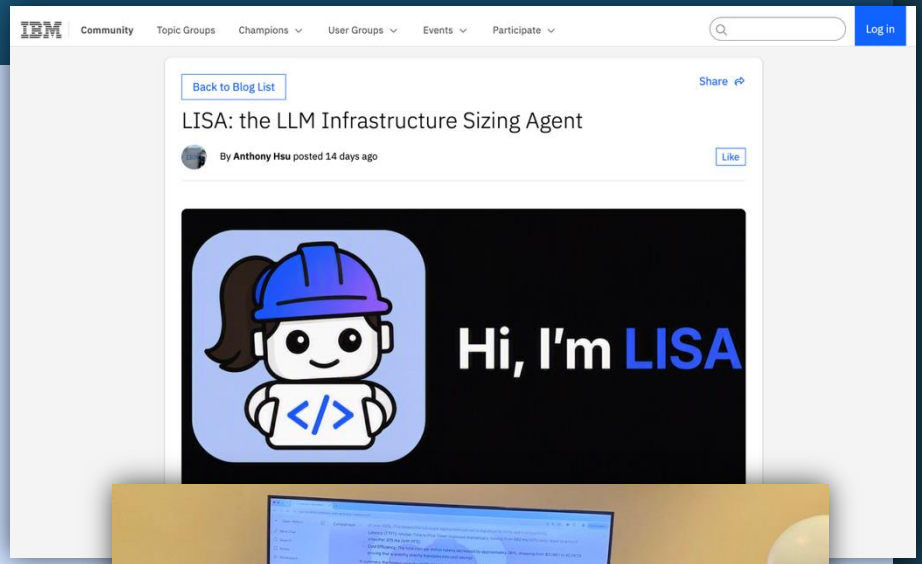
LISA

NA

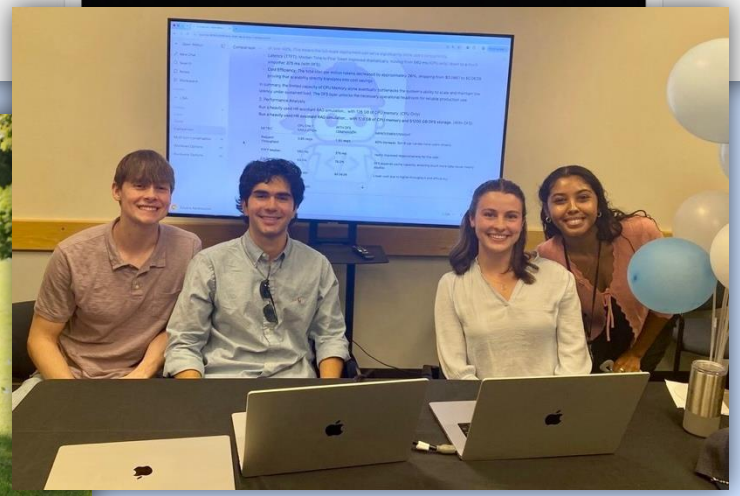
Beyond the project…

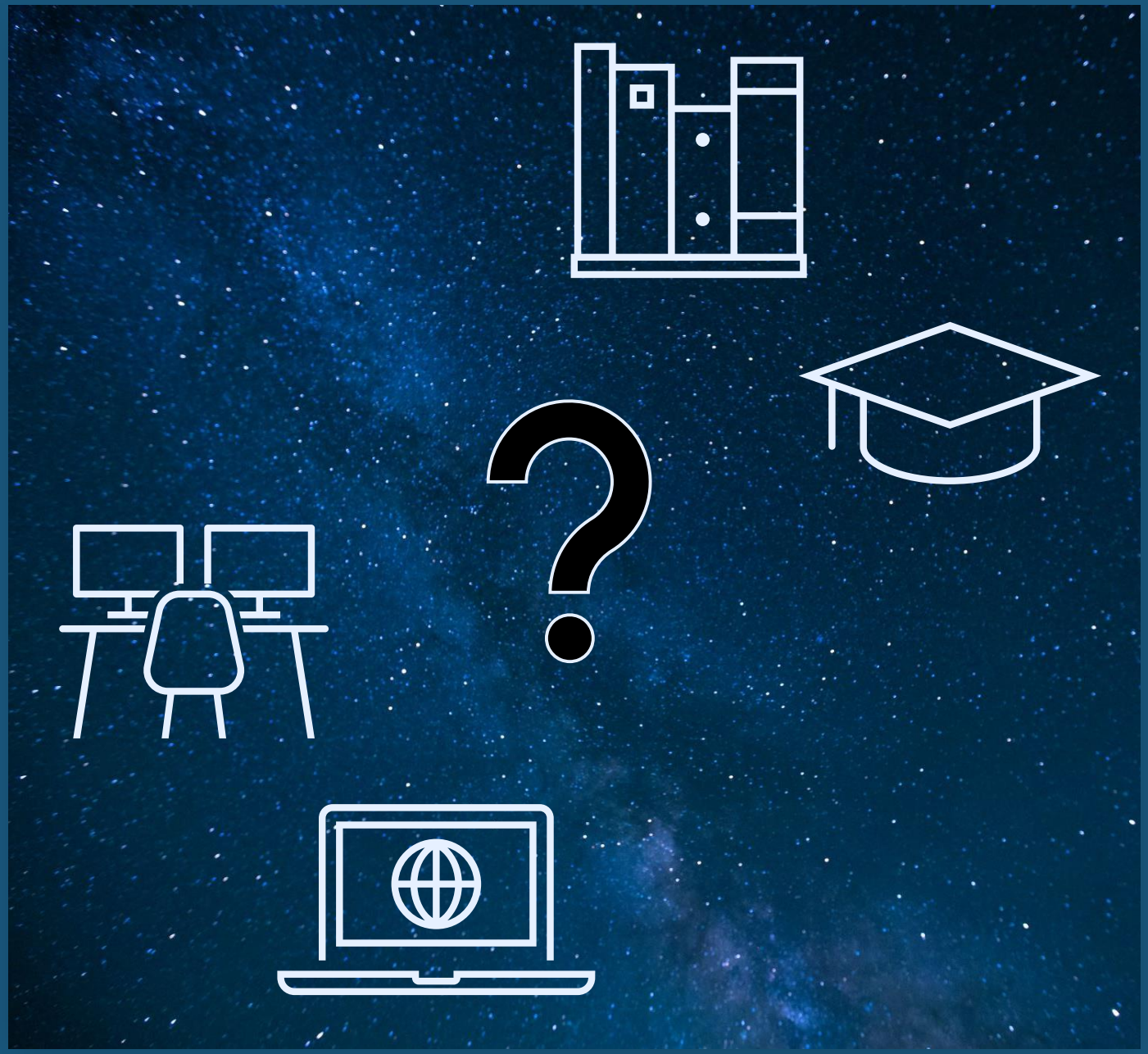
- Technical & pitch presentations
- TechXchange blog
- International collaboration
- Continuous research on cutting edge AI inference technologies











My industry internship has enabled another career path for me

- Agentic programming
- Leading AI technologies
- Industry experience
- Networking

Acknowledgements

These opportunities and accomplishments would not be possible without the support of my mentors and organizations:

- Vincent Hsu - IBM Fellow, VP & CTO IBM Storage
- Khanh Ngo - Senior Technical Staff Member for IBM Storage CTO Office
- Dr. CK Chan - Associate Astronomer and Research Professor
- TIMESTEP (Gurtina, Jen, Rebecca, Graduate student mentors, and more)
- IBM Tucson
- University of Arizona Astronomy & Physics Departments

Thank You

Nayera Abdessalam
nmabdessalam@arizona.edu